End-to-End Sales Analysis

Week 4 Capstone Project - Data Visualization & Analysis Report

Executive summary

This report analyses 700 orders (September 2013 to December 2014) across five customer segments, five countries and six products. It cleans the data, explores it, answers business questions with SQL and Pandas, builds a regression model to predict sales, and ends with recommendations for stakeholders.

Metric	Value
Orders / units sold	700 / 1,125,806
Net sales	$118,726,350
Profit	$16,893,702 (14.2% margin)
Discounts given	$9,205,248 (7.2% of gross sales)
Loss-making orders	58 orders, -$777,321

What the data says

- **Government drives the business:** 44.2% of sales and 67.4% of profit. Government plus Small Business is about 80% of sales.
- **Enterprise loses money:** margin -3.1% (-$614,546 profit). All 58 loss-making orders are Enterprise orders.
- **Discounts erode profit:** margin is 21.9% with no discount but only 9.1% in the High band, yet High and Medium bands already carry 64.1% of sales.
- **Countries are balanced** ($21M-$25M each), so the profit problem is about segments and pricing, not geography.
- **Seasonality:** October and December are the peak months. Like-for-like (Sep-Dec) sales grew 37% in 2014 versus 2013.
- **Prediction:** price, volume and discount band predict sales almost exactly (test R-squared 0.997); without volume the model reaches only 0.55-0.62, so forecasting volume is the key task.

Top recommendations

- Fix Enterprise economics: raise price, cut unit cost or limit discounting so no order is sold below cost.
- Put approval gates on discounts above the Medium band.
- Protect Government, but reduce reliance on one segment for two-thirds of profit.
- Test scaling Channel Partners and Midmarket, which earn high margins on small sales bases.

1. Data and method

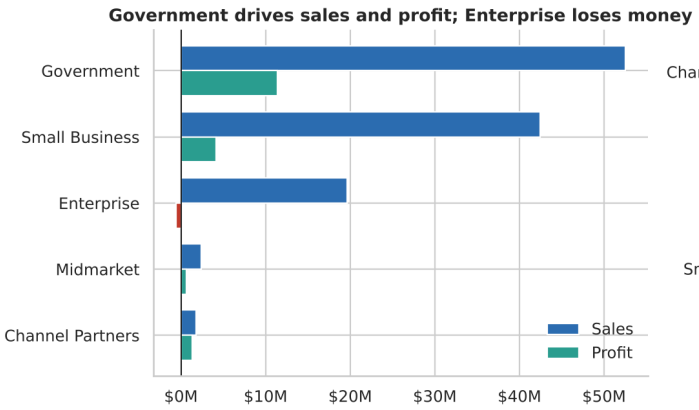
The Google Sheet was supplied as a PDF export. It printed the columns in two halves (Segment to COGS, then Profit to Year), with some text clipped (for example "Governmen", "United Sta"). The table was rebuilt by reading each value by its position on the page and restoring the clipped text from the known categories.

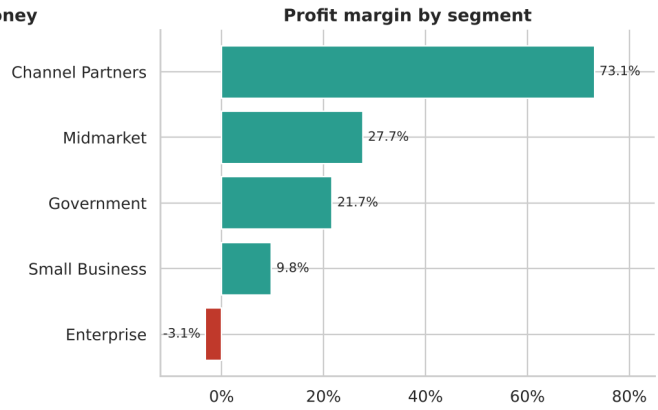
Cleaning and validation

Check	Result / action
Rows and columns	700 rows x 16 source columns (two derived columns added)
Missing values	1 order had blank Sales and COGS; rebuilt as Gross Sales - Discounts and Sales - Profit
Category 'None' (Discount Band)	Kept as a real category, not treated as missing
Dates	Text such as 1-Dec-14 converted to dates; month and year columns match the date
Arithmetic integrity	Units x Price = Gross Sales, Gross - Discounts = Sales, Sales - COGS = Profit: all 700 rows pass
Duplicates	None
New fields	Profit Margin % and Discount %

Tools: Python (pandas, matplotlib, seaborn, scikit-learn, statsmodels), SQLite for SQL queries, an Excel workbook with formula-driven summaries, and a Power BI dashboard built from the same cleaned data.

2. Exploratory analysis

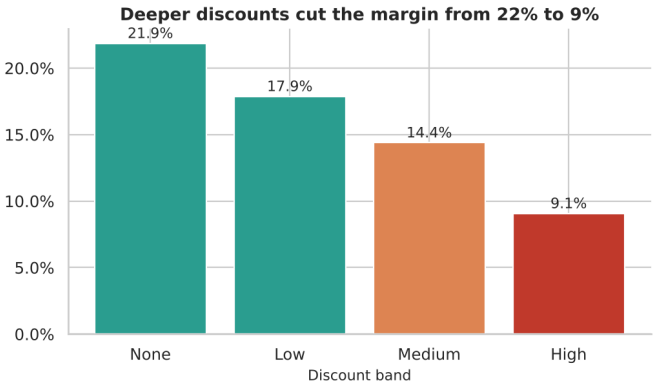


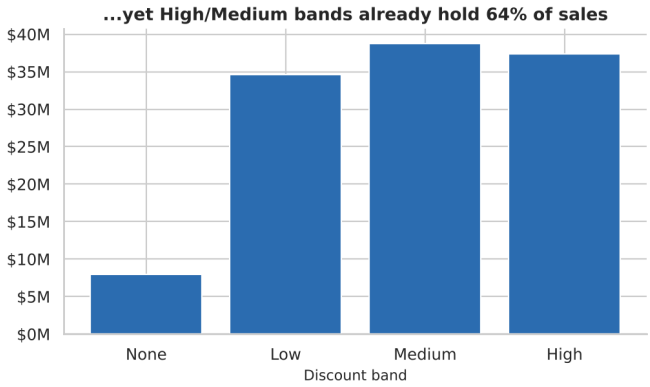


Government and Small Business generate most sales. Enterprise sells $19.6M but loses money; Channel Partners and Midmarket have the best margins on very small sales.

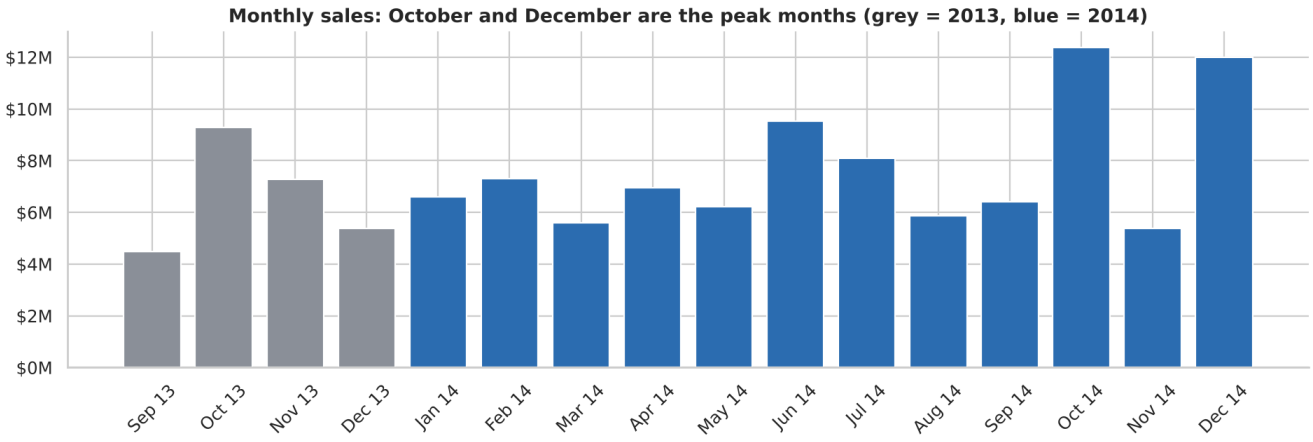
Segment performance (SQL and Pandas agree)

Segment	Orders	Sales	Profit	Margin	Sales share	Profit share
Government	300	$52,504,261	$11,388,173	21.7%	44.2%	67.4%
Small Business	100	$42,427,918	$4,143,168	9.8%	35.7%	24.5%
Enterprise	100	$19,611,694	-$614,546	-3.1%	16.5%	-3.6%
Midmarket	100	$2,381,883	$660,103	27.7%	2.0%	3.9%
Channel Partners	100	$1,800,594	$1,316,803	73.1%	1.5%	7.8%

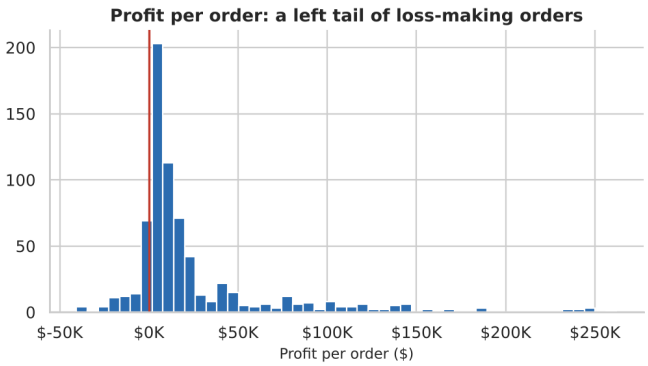


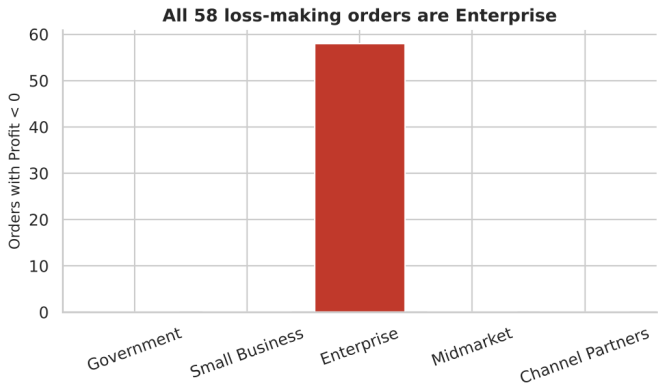


Margins fall steadily as discount depth rises. The High band earns less than half the margin of the no-discount band while carrying almost a third of all sales.

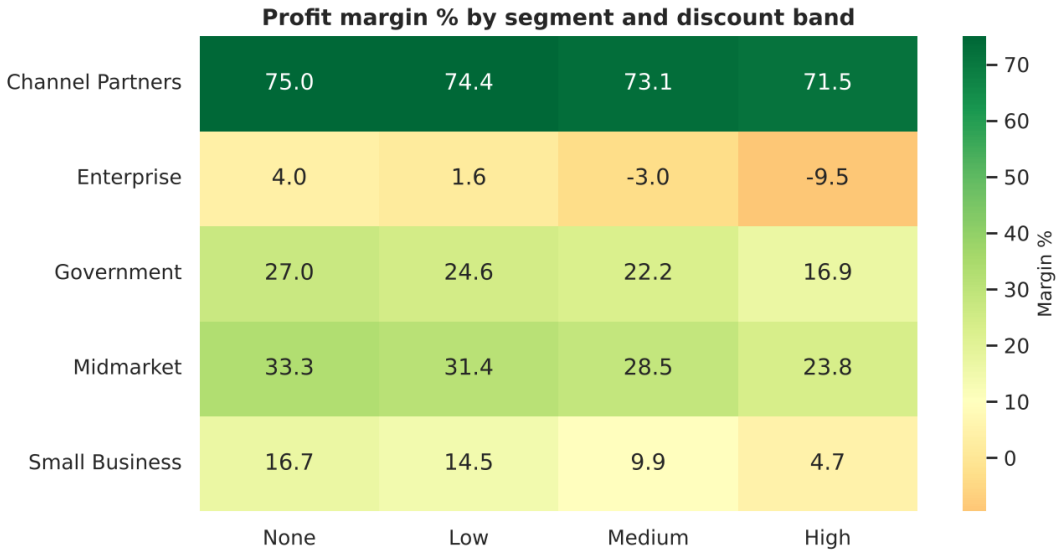


2013 contains only September to December, so the large 2014 total partly reflects extra months. Comparing the same four months gives growth of 37%.





Every one of the 58 orders with negative profit comes from the Enterprise segment, together losing -$777,321. Enterprise cost of goods sold is about 103% of its net sales.



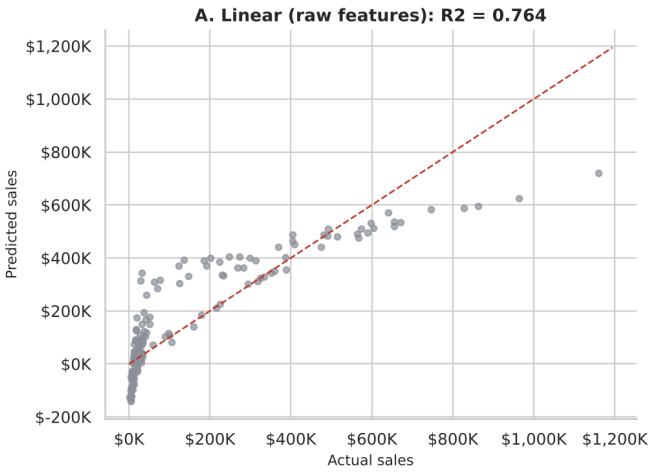
Enterprise is red in every discount band, so the issue is pricing against cost, not only discounting. Government remains healthy even at high discount levels.

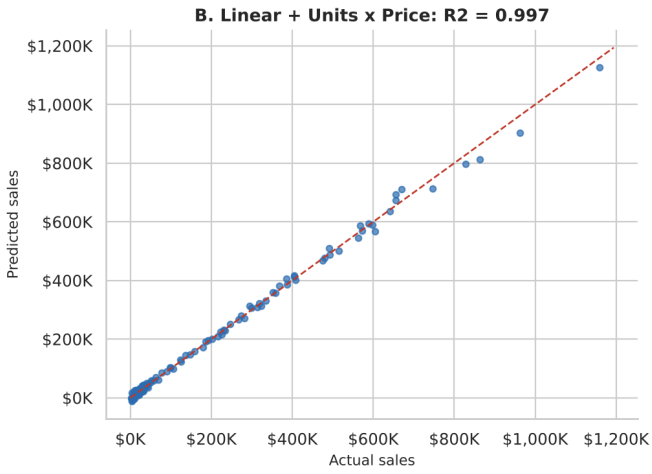
3. Regression model: predicting sales

Target: Sales per order. **Features:** Units Sold, Sale Price, Discount Band, Segment, Country, Product, Month and Year. Gross Sales, Discounts, COGS and Profit were excluded because they are calculated from sales and would leak the answer. Models were scored on a held-out 20% test set and with 5-fold cross-validation.

Model	Test R-sq	CV R-sq	MAE ($)	RMSE ($)
A. Linear (raw features)	0.7643	0.7851	82,877	114,244
B. Linear + Units x Price	0.9971	0.9973	8,804	12,733
C. Random Forest	0.9935	0.9964	6,731	19,020
D. Gradient Boosting	0.9959	0.9976	5,015	15,129

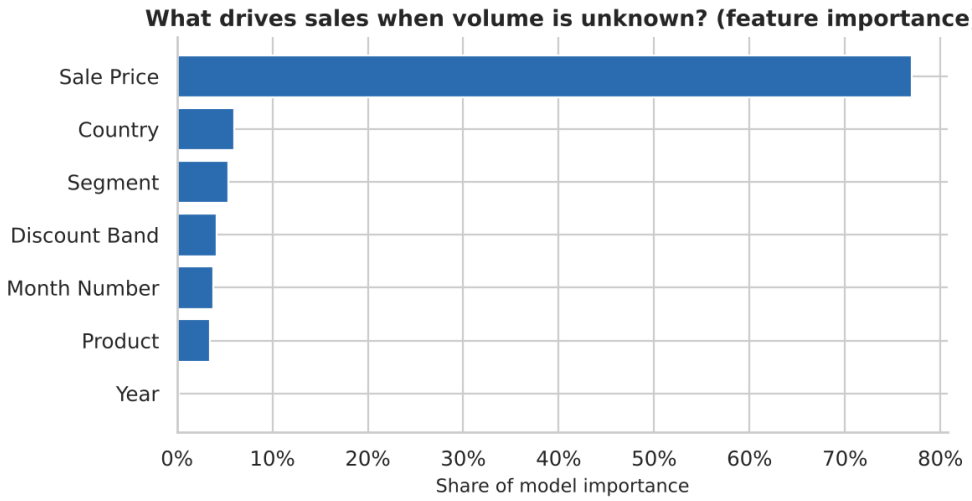
The raw-feature linear model explains 76% of the variation. Adding one engineered feature, Units x Sale Price (list-price revenue), lifts this to 99.7%, matching the tree models while staying fully explainable. It is the recommended model.





Interpretation and limits

- Each $1 of list-price revenue converts to about $0.93 of net sales on average; the High band shifts predictions down by roughly $19,313 per order.
- The model is largely recovering the pricing formula (price x volume less discount), so it is only as good as the volume figure it is given.
- A planning model without volume reaches test R-squared 0.55 (cross-validated 0.62); Sale Price dominates. Forecasting units is therefore the next step.
- Only 16 months and 700 orders are available, so seasonality estimates are based on one year of full data.



4. Recommendations

1. Fix Enterprise economics. Raise list prices, lower unit cost or limit discounts so that no order is sold below cost. Break-even alone would recover about $614,546 of profit.

2. Govern discounting. Require approval for discounts above the Medium band and steer High-band deals toward Low. High-band margin is less than half the no-discount margin.

3. Protect Government, diversify. Government delivers two-thirds of profit; losing a few large deals would hurt. Grow Midmarket and Small Business margins to balance the mix.

4. Test scaling Channel Partners and Midmarket. 73% and 28% margins on small sales bases are attractive; validate that margins hold at higher volume before committing investment.

5. Plan around seasonality. Stock and promote for the October and December peaks.

6. Forecast volume. Build a units forecast by segment and month and feed it into the price x discount model.

Deliverables

- capstone_analysis.ipynb - cleaning, EDA, SQL/Pandas analysis and regression (Python notebook)
- Capstone_Excel_Analysis.xlsx - formula-driven summaries, charts, regression results and calculator (Excel)
- powerbi/ - cleaned CSV, DAX measures and a step-by-step build guide for the Power BI dashboard (.pbix is built in Power BI Desktop)
- Capstone_Final_Report.pdf - this explanation report
- capstone_code.zip - all code and data used